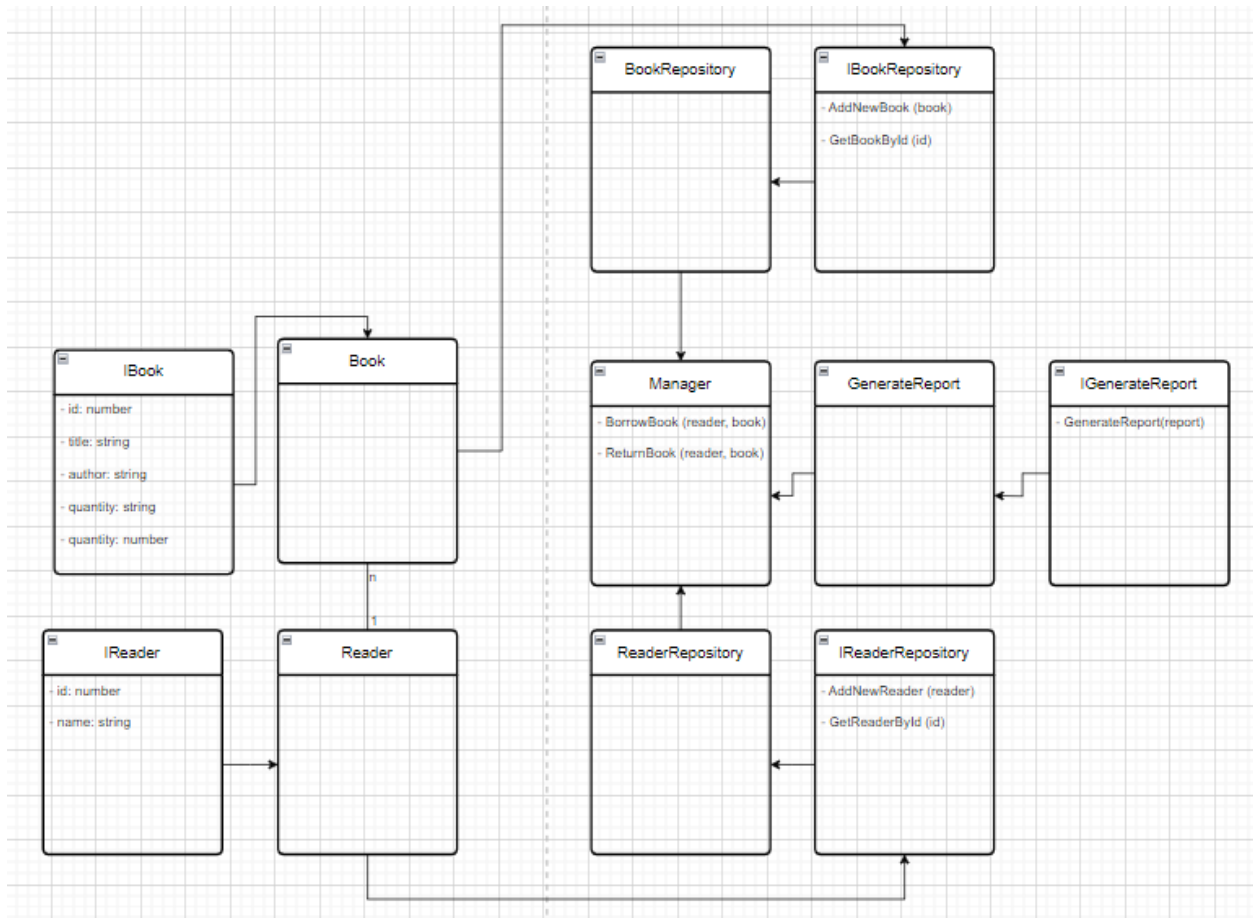




Entities

- **Book:** Represents a book in the library.
- **Reader:** Represents a library reader.

Interfaces (Abstraction Layer)

- **IBook:** Defines the data structure for books.
- **IReader:** Defines the data structure for readers.
- **IBookRepository:** Defines methods for interacting with book data.
- **IReaderRepository:** Defines methods for interacting with reader data.
- **IGenerateReport:** Defines methods for generating reports.

Repositories (Data Access Layer)

- **BookRepository:** Implements IBookRepository, manages book data.
- **ReaderRepository:** Implements IReaderRepository, manages reader data.

Service/Manager (Business Logic Layer)

- **Manager:** Handles book borrowing and returning operations.
- **GenerateReport:** Responsible for generating reports based on IGenerateReport.

Applying SOLID Principles

- **Interface Segregation:** Interfaces are small and separated based on functionality.
- **Dependency Inversion:** The Manager class does not directly depend on repositories but interacts through interfaces.